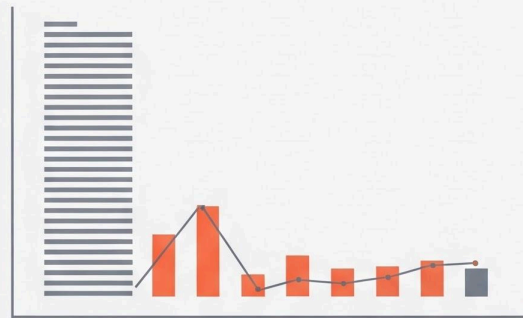




IGN Analytics



Data Analytics Project: **IGN Game Reviews Analysis**

Dataset Used: `ign.csv`

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze game review scores, platform popularity, and genre distribution over time using the `ign.csv` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (● Easy)

Description:

You load the game dataset and quickly look at its contents using the first and last rows and its size. Then, you clean the data by removing an unnecessary column. After that, you check for missing values in important columns and fill them—using the average for scores and the most common value for genres. Finally, you make sure all columns have the correct data types so the dataset is ready for further analysis and visualization.

Task	Description
1.	Load the <code>ign.csv</code> dataset using Pandas.
2.	Display the first 5 rows (<code>head()</code>), last 5 rows (<code>tail()</code>), and the shape of the dataset.
3.	Remove the unnecessary column: <code>"Unnamed: 0"</code> .
4.	Check for missing values in <code>score</code> , <code>genre</code> , and <code>platform</code> .
5.	Handle missing values: Fill the numeric column <code>score</code> with the mean, and the categorical column <code>genre</code> with the mode.
6.	Ensure correct data types: Convert <code>score</code> → float and <code>release_year</code> , <code>release_month</code> , <code>release_day</code> → integer.

Conclusion:

After completing these steps, the dataset becomes clean, consistent, and ready for analysis. Missing values are handled, unnecessary data is removed, and all columns are in the correct format. This ensures that any further analysis, graphs, or insights you generate from the data will be accurate and reliable.

Scenario 2: Line Graph (Score Trend) + Save (● Easy)

Description:

This task analyzes how game scores change over time using a line graph. The data is grouped by release year, and the average score for each year is calculated. These values are then converted into arrays for plotting. A line graph is created with years on the x-axis and average scores on the y-axis. Finally, the graph is labeled properly and saved as an image for future use.

Task	Description
1.	Group data by <code>release_year</code> .
2.	Calculate the average score per year using Pandas.
3.	Convert the resulting average scores and years into NumPy arrays.
4.	Plot a line graph with <code>release_year</code> on the X-axis and average score on the Y-axis.
5.	Add the title: "Average Game Score Over Years" and proper axis labels.
6.	Save the graph: <code>plt.savefig("avg_score_trend.png")</code> .

Conclusion:

The graph shows how game scores change from year to year in an easy way. By looking at it, we can understand if games are getting better, worse, or staying the same over time. It helps us quickly spot trends without checking each value. Overall, it makes the data easier to understand and useful for analysis.

Scenario 3: Filtering + Bar Chart + Save (🟡 Medium)

Description:

This task focuses on comparing the top gaming platforms based on high-rated games. First, only games with scores greater than 7 are selected. Then, the number of such games is counted for each platform. The top 10 platforms are chosen based on these counts. A bar chart is created to visually compare them, and the labels are rotated for better readability. Finally, the chart is saved as an image file.

Task	Description
1.	Filter the dataset to include only games where <code>score > 7</code> .
2.	Count the number of high-rated games per platform.
3.	Select the top 10 platforms using Pandas.
4.	Convert the platform counts into NumPy arrays.
5.	Plot a bar chart with <code>platform</code> on the X-axis and count of games on the Y-axis.
6.	Rotate x-axis labels for readability.
7.	Save the graph: <code>plt.savefig("top_platforms_bar.png")</code> .

Conclusion:

This chart helps us easily see which platforms have the most high-rated games. By focusing only on games with good scores, we get a clearer idea of platform performance. It makes comparison simple and quick. Overall, it shows which platforms are more popular for quality games.

Scenario 4: Aggregation + Pie Chart + Save (🟡 Medium)

Description:

This task analyzes how games are distributed across different genres. First, the total number of games in each genre is counted. Then, the top 5 most common genres are selected. A pie chart is created to show their distribution visually. Percentage labels are added to make it easier to understand each genre's share. Finally, the chart is saved as an image file.

Task	Description
1.	Count the total number of games per <code>genre</code> .
2.	Select the top 5 genres using Pandas.
3.	Prepare labels (genres) and values (counts) for the pie chart.
4.	Plot a pie chart using <code>genre</code> as labels and <code>count</code> as values.
5.	Add percentage labels using <code>autopct</code> (e.g., '%.1f%%').
6.	Save the graph: <code>plt.savefig("genre_distribution.png")</code> .

Conclusion:

This chart helps us clearly see which game genres are the most popular. It makes it easy to compare how much each genre contributes overall. By looking at the percentages, we can quickly understand the distribution. Overall, it gives a simple view of genre popularity.

Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)

Description:

This task performs a deeper analysis of game review data by creating new features and visualizations. Scores are categorized into groups like Excellent, Good, and Average, and editor choices are converted into numeric form. Using NumPy, changes in average scores over the years are calculated. Multiple graphs like line charts, stacked bar charts, and histograms are used to visualize trends and distributions. Finally, insights are generated to understand score patterns and relationships.

Task	Description
1.	Create a new column <code>score_category</code> : "Excellent" for score ≥ 9 , "Good" for $7 \leq \text{score} < 9$, and "Average" for score < 7 .
2.	Convert the <code>editors_choice</code> column: 'Y' $\rightarrow$ 1, 'N' $\rightarrow$ 0.
3.	Use NumPy's <code>np.diff()</code> to calculate the yearly score growth based on the average yearly scores from Scenario 2.
4.	Plot a Line Graph showing the trend of average score per <code>release_year</code> .
5.	Plot a Stacked Bar Chart showing the count of <code>score_category</code> per <code>release_year</code> .
6.	Plot a Histogram showing the distribution of <code>score</code> .
7.	Save the graphs: <code>plt.savefig("score_trend.png"),</code> <code>plt.savefig("score_category_stacked.png"),</code> <code>plt.savefig("score_distribution.png").</code>
8.	Identify: Which years had the highest scores, whether high scores increased over time, and if <code>editors_choice</code> correlates with high scores.

Conclusion:

This analysis helps us understand how game scores change over time and how they are distributed. It shows which years had better scores and how different categories of games are spread. The graphs make it easier to see patterns and trends. Overall, it gives a clear idea about game quality and how editor choices relate to high scores.

Final Conclusion:

Overall, this project helps us understand and analyze game review data step by step. We start by cleaning the dataset and fixing missing values so the data becomes reliable. Then, we use different graphs like line charts, bar charts, and pie charts to understand trends, compare platforms, and see genre distribution. Finally, advanced analysis helps us discover patterns in scores and how they change over time. In simple terms, the whole process turns raw data into meaningful insights, making it easier to understand game quality, popularity, and trends.

What you have learned

I learned how to work with data step by step through different scenarios. First, I understood how to clean and prepare the dataset properly. Then, I learned how to analyze trends over time using a line graph. I also learned how to compare platforms and genres using bar charts and pie charts. Finally, I explored advanced analysis to find patterns and relationships in the data. Overall, these scenarios helped me understand how to analyze and visualize data in a practical way.